STEP 1
Order Entry
ERP sales order

STEP 2
Credit & Release
ERP order status

STEP 3
Pick & Pack
WMS

STEP 4
Carrier Tender
TMS

STEP 5
Pickup & Transit
Carrier EDI 214 / visibility

OUTCOME (Y)
Delivered On Time?
Proof of delivery

▲ Queue: awaiting credit/release

▲ Queue: awaiting wave/pick

▲ Queue: awaiting tender

▲ Queue: awaiting pickup

▲ In transit

Decision window: risk score must be available here to expedite, re-sequence, or re-tender

Limited options

Too late to act

DECISION
Can we commit to the requested date?

QUESTIONS
Was the order placed before the daily cutoff?

Is the requested date feasible?

VARIABLES
✓ Order time of day
`order date`
↳ *ERP order header*

✓ Payment type
`Type`
↳ *ERP order header*

✓ Price, discount, quantity
`Sales…`
↳ *ERP order lines*

✓ Customer segment, market
`Market`
↳ *CRM customer master*

◐ Requested vs. promised date gap
↳ *ERP order header*

◐ Available-to-promise qty
↳ *ERP inventory and ATP*

Primary system: ERP sales order

DECISION
Release the order now or hold it?

QUESTIONS
How complex is the order?

How long has it waited for release?

VARIABLES
✓ Lines in order (derived)
`Order Id`
↳ *ERP order lines*

◐ Order-to-release hours
↳ *ERP status log*

◐ Slack days to promise date
↳ *ERP order header*

! Order status at extract
`Order Status`
↳ *ERP status at extract time*

Primary system: ERP status change log

DECISION
Prioritize this order in the wave?

QUESTIONS
Is the network running behind?

Which facility ships it?

VARIABLES
✓ Regional order load, prior 7 days (derived)
`order date`
↳ *Derived from ERP order history*

◐ Pick backlog at release
↳ *WMS task queue*

◐ Pick and pack timestamps
↳ *WMS scan events*

✗ Shipping facility
↳ *Not recorded*

✗ Shipment weight
↳ *WMS pack station*

Primary system: WMS task records

DECISION
Expedite, switch carrier, or warn the customer?

QUESTIONS
Is the service level achievable?

Is this lane running late?

VARIABLES
✓ Shipping mode
`Shipping Mode`
↳ *ERP order header*

✓ Scheduled shipping days
`Days for shipment (scheduled)`
↳ *ERP order header*

✓ Lane: mode x region (derived)
`Order Region`
↳ *Derived from ERP order and region*

◐ Carrier and tender acceptance
↳ *TMS tender history*

◐ Hours from tender to pickup
↳ *TMS appointments*

Primary system: TMS tender records

DECISION
Intervene while in transit?

QUESTIONS
Did it ship on time?

Is the ETA slipping?

VARIABLES
! Actual ship date
`shipping date`
↳ *WMS ship confirmation*

! Actual shipping days
`Days for shipping (real)`
↳ *WMS ship confirmation*

✗ Milestone ETA variance
↳ *Carrier portal or EDI 214*

✗ Weather and congestion
↳ *External data feed*

Primary system: carrier milestones

TARGET
Shipment late (1) or not (0)

VARIABLE
✓ Target
`Late_delivery_risk`
↳ *Proof of delivery*

Known only after the outcome. Exclude.

! Label restated
`Delivery Status`
↳ *Derived from proof of delivery*

! Actual days define the label

Population rule. 7,754 canceled shipments have no delivery outcome and are removed.

Identifiers and PII excluded: names, email, password, street, IDs

✓ In the dataset (or derived from it) ◐ Captured elsewhere in the organization, missing from dataset

✗ Not captured ! Timing violation (post-outcome) ↳ *Where the data stream lives* ▲ Wait / queue

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predictor variables available

4
post-outcome fields flagged

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timing and queue variables